

GIT 3: Quiver Representations

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Outline

1. What is a quiver representation?
 - (a) Quiver
 - (b) Representations, kQ -modules and dimension vector
 - (c) A word about relations
2. GIT problem
 - (a) Space $\mathcal{R}(Q, \underline{d})$ and $\mathrm{GL}(\underline{d})$ group action.
 - (b) G -linearized line bundles from functionals on Grothendieck group.
 - (c) Sketch of Hilbert-Mumford proof/analysis
 - (d) Definition of θ -stability a la King.
 - (e) Moduli functor, fact that it is a coarse moduli space up to S -equivalence.
3. An Example
 - (a) Generalized Kronecker quiver.
 - (b) Beilinson quiver.